

Datatypes

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Mutable Immutable | | Lists Tuple

Immutable Data Types: Cannot be changed after creation (If you modify them, a new object is created.)

Types:

int → 10 float → 3.14 complex → 2 + 3j bool → True / False str (String) → "Hello" tuple → (1, 2, 3)

x = 10 x = x + 5 # New object created

s = "Hello" s[0] = "h" # Error (String is immutable)

2. Mutable Data Types

Can be changed after creation

Types: list → [1, 2, 3] dict (Dictionary) → {"name": "Ali"}

List (Important Mutable Type) A list is an ordered collection of items.

Example: numbers = [1, 2, 3] numbers.append(4) # Modify list print(numbers)

✓ Output: [1, 2, 3, 4]

You can: Add items → append() Remove items → remove() Change items → numbers[0] = 10

Slicing

-7-6-5-4-3-2-1 iotclass 01234567

print(s[:]) | | starting stop index order (inclusive) (exclusive)

```
print(s[1: ]) otclass
```

```
print(s[ :3]) iot
```

```
print(s[2 :-2]) Tcla
```

```
print(s[:6]) #starting to fifth
```

```
In [ ]: s='''IoTclass'''          #Slicing
        print(s[:])
```

```
print(s[1:])
print(s[:6]) #starting to fifth
print(s[-5:-1])
print(s[2 :-2])
```

IoTclass

oTclass

IoTcla

clas

Tcla

Reversed the string

```
In [1]: text = "hello"
        reversed_text = text[::-1]
        print(reversed_text)
```

olleh

1. Concatenation
2. upper, lower case
3. Slicing
4. endswith(' ')
5. capatilize()
6. replace(' ','')
7. find()
8. count(' ')

```
In [8]: s='''IoTclass''' #concatenation
        t="Welcome to "
        print(t+s)

        print(s.upper()) #upper case

        print(s.lower()) #lower case

        print(s[0])
```

Welcome to IoTclass

IOTCLASS

iotclass

I

```
In [14]: s='''IoTclass'''
         t="Welcome to "
         print(s.endswith('class'))
```

```
print(s.capitalize())
```

True

IoTclass

```
In [ ]: s='''IoTclass'''
        t="Welcome to "
        print(t.replace('e','a'))
```

Walcoma to

```
In [ ]: s='''IoTclass'''
        t="students! Welcome to IoT class"
        print(t.find('IoT'))           #index of first occurrence
```

21

```
In [1]: s='''IoTclass'''
        t="IoT students! Welcome to IoT class"
        print(t.count('IoT'))          #number of occurrence
```

2

```
In [ ]:
```